

- Document Control
- System Name: Application
- Version: 1.0
- Author: Data Engineering Team
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- Status: Draft
- System Overview
 - The Application delivers an automated ETL pipeline leveraging Delta Live Tables to ingest, transform, clean, aggregate, and maintain SCD Type 2 history for CSV-sourced customer and order data. It automates structured data onboarding, enforces schema and data quality, and supports spend and transaction history analysis.
- User Roles
 - Data Engineer
 - Monitors and maintains ETL jobs
 - Configures pipeline schedules
 - Business Analyst
 - Reviews and analyzes output data (not involved in technical implementation)
 - Compliance Officer
 - Ensures privacy and audit requirements are satisfied
- Functional Scope
 - Source data ingestion from CSV files
 - Schema enforcement on customer and order tables
 - Data transformation, cleansing, and aggregation logic
 - Derivation and calculation of new columns (e.g., TotalAmount)
 - Automated table creation and management
 - SCD Type 2 tracking on ordersummary
 - Aggregation of customer spend at required grains
 - End-to-end process orchestration via Delta Live Tables
- Functional Requirements

FR-ETL-001 — Load Source Customer Data

- Description: System shall read customer CSV files and load records into designated delta table storage.
- Preconditions: CSV files are available and accessible at the specified source path.
- Main Flow:

1. Monitor source file location for new/updated customer CSV files
2. Validate file format and schema compatibility
3. Load data into /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata delta table
 - Alternate Flows: Log and alert on missing/invalid files; skip records not matching schema
 - Acceptance Criteria: All valid records from source CSV are loaded and persisted as delta table rows

FR-ETL-002 — Load Source Order Data

- Description: System shall read order CSV files and load records into orderdata delta table.
- Preconditions: Order CSV files are present in designated source path.
- Main Flow:
 1. Monitor source for new/updated order CSV files
 2. Perform file structure and schema validation
 3. Load valid records to /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
 - Alternate Flows: Skip files with mismatched schema; error out incomplete loads with notification
 - Acceptance Criteria: All valid order records are loaded into orderdata table

FR-ETL-003 — Customer Table Schema Enforcement

- Description: System shall enforce schema: CustId, Name, EmailId, Region on customer table.
- Preconditions: Schema mapping defined in pipeline configuration
- Main Flow:
 1. Parse incoming data
 2. Validate against expected fields
 3. Enforce types and reject inconsistent data
 - Alternate Flows: Records failing schema validation routed to error log
 - Acceptance Criteria: Customer table contains only records matching defined schema

FR-ETL-004 — Order Table Schema Enforcement

- Description: System shall enforce schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table.
- Preconditions: Defined table schema
- Main Flow:

1. Ingest order data
2. Validate all record fields and data types
3. Apply rejection for records not matching schema
 - Alternate Flows: Capture invalid records in error logs
 - Acceptance Criteria: Only valid records persist in order table

FR-ETL-005 — Calculate TotalAmount Field

- Description: System shall add and calculate TotalAmount field in order data as (PricePerUnit * Qty).
- Preconditions: orderdata records have valid PricePerUnit and Qty values
- Main Flow:
 1. For each orderdata record, compute TotalAmount
 2. Append TotalAmount as new column
 3. Persist augmented record
- Alternate Flows: If either field is null/invalid, label record for remediation
- Acceptance Criteria: Every order data row has correct TotalAmount value

FR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: System shall remove records with null or duplicate values from customer and order tables.
- Preconditions: Tables populated with source data
- Main Flow:
 1. Scan for null or duplicate records post-load
 2. Remove offending records
 3. Log number and details of removals
- Alternate Flows: Notify if duplicate/null rate exceeds threshold
- Acceptance Criteria: Post-processing, tables contain no null or duplicate records

FR-ETL-007 — Create Ordersummary Table

- Description: System shall create the ordersummary table if it does not already exist.
- Preconditions: Permissions to create tables in target schema
- Main Flow:
 1. Check existence in gen_ai_poc_databrickscoe.sdmc_wizard

2. Create table with defined schema if missing

- Alternate Flows: Alert on insufficient permissions
- Acceptance Criteria: Ordersummary table exists with specified schema ready for data

FR-ETL-008 — SCD Type 2 on Ordersummary

- Description: System shall maintain historical changes on ordersummary using SCD Type 2 methods.
- Preconditions: SCD Type 2 logic and tracking fields configured
- Main Flow:

1. Detect changes from customer table

2. Update existing ordersummary records (setting end date, inactive status)

3. Insert new record version for changed entities

- Alternate Flows: Log any SCD update errors for remediation
- Acceptance Criteria: All changes are tracked and auditable per SCD Type 2 conventions

FR-ETL-009 — Create CustomerAggregateSpend Table

- Description: System shall create customeraggregatespend table with Name, TotalAmount, Date columns if not exists.
- Preconditions: Correct access on target database/schema
- Main Flow:

1. Check for table existence

2. Define and create table if not present

- Alternate Flows: Alert if unable to create table
- Acceptance Criteria: customeraggregatespend table ready with defined schema

FR-ETL-010 — Aggregate Customer Spend

- Description: System shall aggregate TotalAmount grouped by Name and Date from ordersummary and store in customeraggregatespend.
- Preconditions: Populated ordersummary data
- Main Flow:

1. Query ordersummary for required fields

2. Group by Name, Date, summarize TotalAmount

3. Insert results into customeraggregatespend

- Alternate Flows: Reprocess if aggregation errors encountered
- Acceptance Criteria: Aggregated records match expected totals per customer and date

FR-ETL-011 — Delta Live Tables Implementation

- Description: System shall orchestrate the ETL process using Delta Live Tables end-to-end capabilities.
- Preconditions: Delta Live Tables enabled and configured
- Main Flow:

1. Configure pipeline DAG to sequence ETL jobs
2. Monitor pipeline execution and log status
3. Alert on any execution failures

- Alternate Flows: Retry or escalate failed jobs
- Acceptance Criteria: All ETL tasks complete successfully via DLT
- Business Rules
 - All required source files must be present and schema-conformant before data loading
 - Data integrity and referential constraints must be enforced throughout the pipeline
 - SCD Type 2 logic applies to all customer attribute changes for ordersummary
- Error Handling
 - Invalid file formats trigger error logs and notifications
 - Loading errors result in stopping load process and alerting stakeholders
 - All records failing schema or data quality checks are isolated and logged for review
- Dependencies
 - Availability of reliably sourced CSV data
 - Operational Databricks/Delta Lake and Delta Live Tables environment
 - Pipeline configuration, access rights, database/schema pre-creation
- Acceptance Criteria
 - All valid source records are loaded into target tables per schedule
 - Data quality and integrity enforced
 - Historical and aggregate tables populated as per business needs
 - Full pipeline runs complete without critical errors or data loss